Online Resource 1: Exact slack certificates for protecting original-core maxima

Maxwell Flanagan

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Supplement to Sharp subsidies for protecting core maxima in three-sided assignment markets

No affiliation. City/country and corresponding email remain to be supplied.

1 Scope and notation

This supplement accompanies the complete three-sided proof in the main paper. It retains the exact slack identities from Section 7 of the October 7 compendium; it introduces no new theorem. The broader higher-sector, flow, transportation and ancillary appendices remain in that preserved compendium and are not part of this article's claimed contribution.

There are three finite sectors of equal size n, nonnegative pair-additive full-team values, a nonempty original core and total matching worth $T > 0$. Only full teams produce value. For each player (a, j), $H_{a,j}$ is its separately attained maximum over the entire original core. Relabel an optimal perfect matching diagonally; its team values d_j sum to T. If $u_{a,j}$ denotes the sum of the two diagonal pair values incident to (a, j), then

$$\sum_{a,j} u_{a,j} = 2T, \qquad 0 \le H_{a,j} \le d_j, \qquad \delta_{a,j} = d_j - H_{a,j} \ge 0.$$

For a player $i = (a, j)$, write $u_i = u_{a,j}$ and $H_i = H_{a,j}$. A fractional partial matching has nonnegative weights λ_e, player loads $p_i = \sum_{e \ni i} \lambda_e \le 1$ and residual weights $q_i = 1 - p_i$. Define

$$k = n - \sum_e \lambda_e, \quad W = \sum_e \lambda_e v(e), \quad S = \sum_i q_i H_i, \quad V = 3W + S, \quad \beta = \sum_i q_i u_i, \quad C_2 = \sum_i q_i^2 u_i,$$

$$E = \sum_{a,j} \Big(\sum_{b \ne a} q_{b,j} \Big) \delta_{a,j} \ge 0.$$

The residual mass in each sector is k. For a team $e = ((1, j_1), (2, j_2), (3, j_3))$ and every ordered sector permutation (a, b, c), the main paper proves the six separately witnessed comparisons

$$H_{c,j_c} - H_{c,j_b} \ge v(e) - d_{j_b} - u_{a,j_a}. \tag{1}$$

It also proves, for $k > 0$ and every player i,

$$kH_i \le T - W + (k - q_i)u_i.$$

These inequalities use different attaining core points where necessary; no simultaneous attainment is assumed. They imply the nonnegative slacks below.

2 An exact slack certificate

For a team e and a sector permutation (a, b, c), let

$$\sigma_{e;a,b,c} = d_{j_b} + u_{a,j_a} + H_{c,j_c} - H_{c,j_b} - v(e) \geq 0, \qquad L = \sum_e \lambda_e \sum_{(a,b,c)} \sigma_{e;a,b,c}.$$

When $k > 0$, define

$$F = \sum_i q_i \big[T - W + (k - q_i)u_i - kH_i\big] \geq 0.$$

The two aggregations give the exact equalities

$$E + L = 10T - 2\beta - 2V, \qquad F = 3kT + k\beta - C_2 - kV.$$

Eliminating β yields the nonnegative slack certificate

$$2k(4T - V) = \frac{k}{2}(E + L) + C_2 + F. \tag{2}$$

For an integral partial matching, $C_2 = \beta$ and the corresponding identity is

$$(8k - 5)T - (2k - 1)V = \frac{k - 1}{2}(E + L) + F. \tag{3}$$

Both identities follow by substitution; their signs come from the original-core arguments in the main paper, rather than a numerical feasibility test. On the special partial matchings of the main paper's sharpness section, $E = L = F = 0$, consistent with equality in the integral bound.

The same calculation has a useful necessary-condition interpretation. For an integral P with $k \geq 1$, suppose numbers $h_{a,j}$ obey the diagonal caps $h_{a,j} \leq d_j$, the six comparisons (1) for every $e \in P$ with h replacing H, and $kh_i \leq T - v(P) + (k - 1)u_i$ for every unmatched player. Then the proof of the main paper's integral residual-count theorem gives the same bound with $\sum_{i \in U} h_i$ replacing $\sum_{i \in U} H_i$. Thus these necessary conditions already imply the desired obstruction bound; they need not characterize which vectors are actual core maxima.

3 What the certificate does and does not prove

For $k > 0$, the first identity exhibits $V \leq 4T$. The strict finite-market argument still requires the main paper's cases: if $C_2 > 0$, the inequality is strict; if $C_2 = 0$, then $\beta = 0$ and the residual bound gives $V \leq 3T$; if $k = 0$, all residual weights vanish and $W \leq T$. Compactness of the fractional matching polytope lets strictness be applied at its maximizing point. Thus every fixed finite market has $s(M) < T/3$.

The integral refinement uses $C_2 = \beta$ and an integer $k \geq 1$. It bounds an integral obstruction $J(M, P)$, not the full fractional optimum at a fixed market size. The main paper includes a fractional packing with residual mass $k = 1$ and positive obstruction, ruling out that mistaken extension.

The family with parameter $\ell \geq 1$ has $n = 2\ell + 1$, $T = 3(2\ell + 1)$, all $H_i = 3$ and $s(M) = 2\ell$. Consequently $s(M)/T \uparrow 1/3$ without attaining it. It also attains the integral bound at $k = \ell + 1$. This is the current paper's family, not the different triangle family used to bound protection within the original core in the Acyclicity companion.

4 Exact checks and reproducibility

Online Resource 2 contains standard-library Python checks and frozen rational inputs. Run from a fresh extracted copy, choosing a new output filename:

```
python3 -B -O verification/verify.py -output fresh-check.json
```

The verifier checks source/input hashes before running the workers and keeps its mathematical requirements active when Python assertions are disabled. The fresh 17 October run under Python 3.10.14 passed:

- all 27 ordered triples of labels over three symbols, covering all five label-equality types; four per-label polynomial identities; two global slack identities; and 28 deliberately corrupted symbolic identities rejected;
- 48 rational fractional-packing cases, four enriched necessary-condition points, two rejected relaxed false points and one fractional-scope example;
- all separately attaining ideal witnesses and the matching subsidy allocation for $\ell = 1, \ldots, 12$: 3,603,804 full-team inequalities and 12 deliberately damaged allocations rejected.

The enriched points are diagnostic necessary-condition data, not independently certified market realizations. These finite checks corroborate the conventional proof; they do not establish the all-size theorem or historical novelty.

Historical review used an earlier source snapshot. The dated package binds that review, its documented corrections, the unchanged mathematical blocks, the current source/PDFs and the fresh check report. The supplement's notation recap and cross-document links received a focused consistency review, not a new claim of complete independent proof verification. Human specialist endorsement is not presumed.